From Purchases to Communities:

How User Activity Shapes Product Networks Built from Purchase Data

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Abstract

Product networks built from purchase data link two products whenever the same customer buys both. A customer who buys d products contributes $d(d-1)/2$ links, so very active customers can dominate the network, and a common response is to downweight them. We ask when this helps. In a simulation with known product categories, we compare raw co-purchase counts, Jaccard similarity, inverse-activity weighting, and a normalized SVD of the user–product matrix while varying how unevenly activity is spread, whether active users have category preferences, and how strong those preferences are. When active users focus on one category they carry most of the signal, and downweighting them is costly: with the most concentrated activity, weighting raises the misclustering rate from 13% to 49%. When active users buy indiscriminately and preferences are weak, no method recovers the categories; with stronger preferences, weighting lowers the error by up to 52 percentage points. A second design that holds informative purchases fixed confirms this: added preference-free heavy buyers drive raw counts to chance while weighting stays near its baseline. Activity is not a proxy for noise; whether to downweight it depends on what active users buy.

1 Introduction

Purchase records form a bipartite network between users and products. A standard way to analyse such data is to *project* it onto the products, connecting two products when they share buyers, and then to cluster the resulting network (von Luxburg, 2007; Ke, 2026). The clusters describe groups of products that tend to be bought by the same people.

The projection has a well-known weakness: a user who buys d distinct products creates $d(d-1)/2$ product pairs, so a handful of heavy buyers can supply most of the edge weight. The usual remedy, going back to the collaboration networks of Newman (2001), is to give each user's pairs weight $1/(d-1)$, which makes a user's total contribution grow only linearly.

Whether this is a good idea depends on what active users buy, which the weighting ignores (Figure 1). If a heavy buyer spreads purchases across categories, most of their links cross category boundaries and downweighting removes noise; if the same buyer concentrated on one category, downweighting would discard the strongest evidence in the data. We study this tension with three questions:

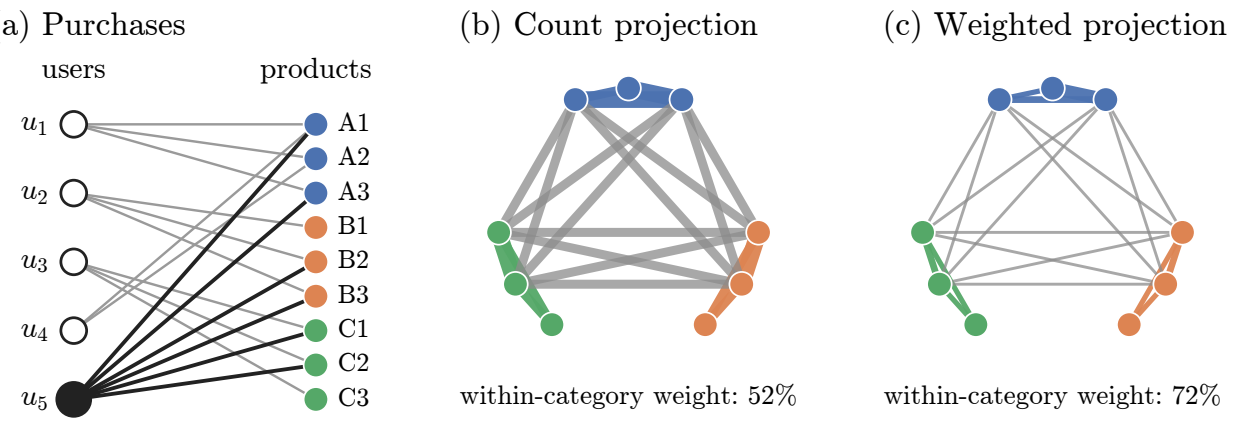


Figure 1: A toy example. Users u_1–u_4 buy within a single category; u_5 buys six products across all three. (b) Under counts, every pair created by u_5 has full weight and cross-category edges (grey) dominate. (c) Inverse-activity weighting gives each of u_5's pairs weight 1/5, and the within-category share of edge weight rises from 52% to 72%. If u_5 instead bought six products from one category, the same weighting would shrink useful edges.

Q1. How do activity heterogeneity and the breadth of active users' preferences interact with the choice of projection weights?

Q2. When does inverse-activity weighting improve on raw counts and Jaccard similarity?

Q3. How do the projections compare with a normalized SVD applied directly to the user–product matrix?

We answer these questions with a simulation in which product categories are known and expected edge weights can be computed exactly. Making heavy buyers indiscriminate both adds noise and, if the amount of data is held fixed, removes informative purchases, so we use two complementary designs: one holds total purchases fixed and one holds the informative purchases fixed. All methods are applied to the same data sets.

Related work. Our $1/(d-1)$ weight follows Newman (2001). Zhou et al. (2007) also discount active users in a resource-allocation projection for recommendation. The SVD baseline is motivated by bipartite spectral co-clustering (Dhillon, 2001) and its extension to directed graphs (Rohe et al., 2016), though we do not implement either algorithm exactly.

2 Model and methods

2.1 A simple purchase model

Let $B \in \{0,1\}^{n\times p}$ record purchases, with $B_{ui} = 1$ if user u bought product i. We use $n = 1{,}000$ users and $p = 300$ products divided into $K = 3$ categories of $s = p/K = 100$ products. Each user has a preferred category, drawn uniformly at random, and buys d_u distinct products one at a time without replacement. At each draw, a remaining product in the preferred category has weight $r_u \geq 1$ and every other product has weight 1. With $r_u = 2$,

about half of a user's purchases fall in the preferred category; with $r_u = 1$ the user has no preference at all.

2.2 Three product networks

Let $S_{ij} = \sum_u B_{ui}B_{uj}$ be the number of shared buyers of products $i \neq j$ and $b_i = \sum_u B_{ui}$ the number of buyers of i. We compare

$$W^{\text{count}}_{ij} = S_{ij}, \qquad W^{\text{jac}}_{ij} = \frac{S_{ij}}{b_i + b_j - S_{ij}}, \qquad W^{\text{wtd}}_{ij} = \sum_{u:\, d_u \geq 2} \frac{B_{ui}B_{uj}}{d_u - 1}.$$

All diagonals are set to zero and no edges are thresholded. For each network W with degree matrix D, we take the eigenvectors of $D^{-1/2}WD^{-1/2}$ belonging to the K largest eigenvalues, normalize each row of the resulting embedding, and run k-means with 20 random starts.

As a fourth method we skip the projection. We form $M = D_u^{-1/2}BD_p^{-1/2}$, where D_u and D_p hold user and product degrees, and cluster the products using the top K right singular vectors of M, with the same row normalization and k-means step. Since

$$M^\top M = D_p^{-1/2}\, B^\top D_u^{-1} B\, D_p^{-1/2}, \tag{1}$$

this baseline is itself a projection that weights each user by $1/d_u$, close to the $1/(d_u - 1)$ of Weighted. The main differences are that it keeps the diagonal of $M^\top M$ and that users with a single purchase still enter it.

2.3 Expected edge weights and user influence

Because products within a category are exchangeable, the probability that user u buys a particular pair of distinct products depends only on whether the two products share a category. Writing C_{uk} for the number of products u buys from category k,

$$q_u^{\text{same}} = \frac{\mathbb{E}[C_{uk}(C_{uk} - 1)]}{s(s-1)}, \qquad q_u^{\text{diff}} = \frac{\mathbb{E}[C_{uk}C_{ul}]}{s^2} \quad (k \neq l).$$

Summing q_u over users gives the expected Count weight of a within- or between-category pair; dividing each term by $d_u - 1$ gives the Weighted expectation. We compute the required moments exactly with a short dynamic program over the sequential sampling scheme (Section A). For Jaccard we use the ratio of expected shared buyers to expected union size, an approximation that agrees with the simulated means to within 0.23% in every condition (Section B).

Two facts follow directly. A user with $r_u = 1$ has $q_u^{\text{same}} = q_u^{\text{diff}}$ and adds no expected within-minus-between contrast, and a user with $d_u \geq 2$ contributes total weight $d_u(d_u - 1)/2$ under Count but only $d_u/2$ under Weighted. Figure 2(b) shows what this means in our design. With focused users and the most concentrated activity, the top 20% of users supply 99.6% of the expected Count contrast; weighting reduces their share to 84.0%. Shares measured in the simulated data agree to within 0.2 percentage points.

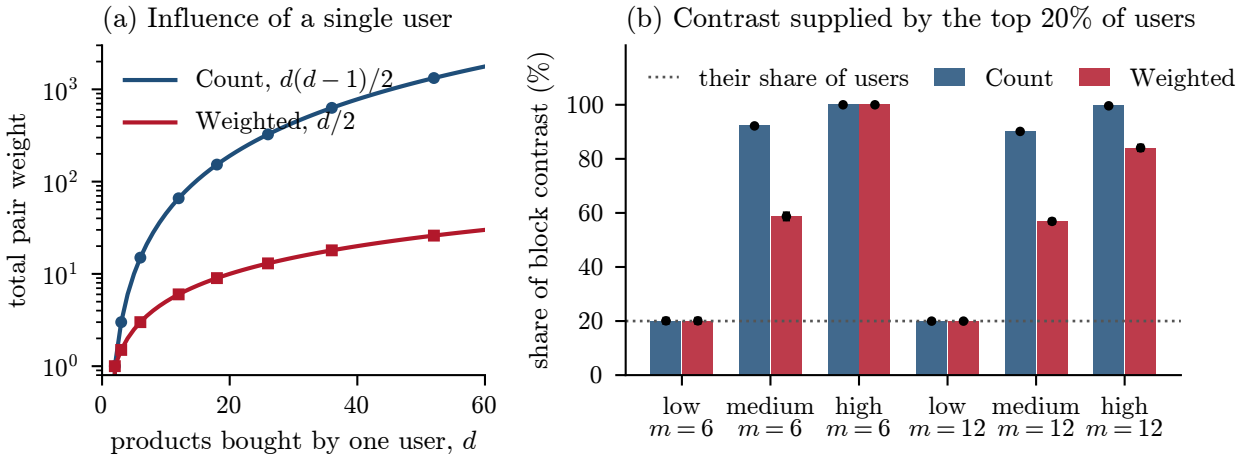


Figure 2: (a) Total pair weight contributed by one user who buys d products; markers show the purchase counts used in our design. (b) Share of the expected within-minus-between contrast supplied by the 20% most active users when all users are focused ($r = 2$). Bars are exact expectations; black points are means over 100 simulated data sets with 95% intervals. Under low heterogeneity the "active" 20% are a random subset of equally active users.

3 Experimental design

Turning the most active users into indiscriminate buyers changes two things. It adds heavy buyers whose pairs carry no category information, and, if the amount of data is held fixed, it takes purchases away from the users who do carry information. No single design can hold both total purchases and informative purchases fixed while doing so, so we use two complementary designs. The *matched-total* design fixes the total number of purchases and asks whether concentrating a fixed amount of activity in a few users helps or hurts each method. The *fixed-information* design fixes the informative users' purchases and adds preference-free heavy buyers on top, which isolates the effect of background noise. Every condition in both designs has 100 independent data sets.

3.1 Matched-total design

We vary three factors (Table 1). *Density* sets the mean number of purchases per user, $m \in \{6, 12\}$. *Activity heterogeneity* determines how those purchases are spread: equally, or with 20% of users buying three or more times the mean. Total purchases are identical across heterogeneity levels, so only their concentration changes. *Preference breadth* is either *focused*, where every user has $r_u = r$, or *broad active*, where the 20% most active users have $r_u = 1$ and the rest keep $r_u = r$. Activity is assigned independently of the preferred category. Finally, the *preference strength* $r \in \{2, 3, 4\}$ sets how strongly focused users favour their category. Crossing the four factors gives 36 conditions.

The settings were fixed after a pilot of 10 data sets per condition. Following our proposal, we required at least two conditions with a median error between 5% and 40% for some method; four met it at $r = 2$, so we kept the initial design, without regard to which method performed best. We first ran the $r = 2$ grid and then the same grid at $r = 3$ and $r = 4$, the

Table 1: Purchase counts per user. Within each density, total purchases are the same in every column.

Mean purchases m	Low heterogeneity	Medium heterogeneity	High heterogeneity
6	all users buy 6	80% buy 3, 20% buy 18	80% buy 1, 20% buy 26
12	all users buy 12	80% buy 6, 20% buy 36	80% buy 2, 20% buy 52

upper value in our proposal, with identical code and fresh seeds.

3.2 Fixed-information design

We fix 1,000 informative users in one of two settings ($m = 12$, $r = 2$ or $m = 6$, $r = 3$), then add $A = 50$ or 200 users with no category preference who each buy D products, with $D/m \in \{1, 2, 4, 8\}$, plus one baseline with no added users. This gives 18 conditions. Within each replicate the informative purchases are identical across conditions, and the added users and their purchases are nested, so comparisons across conditions isolate the background. Unlike the matched-total design, total purchases grow as background is added. We added this design after the matched-total results at $r = 2$ showed that the two effects were confounded.

3.3 Evaluation

Our primary measure is the *misclustering rate*: the fraction of the 300 products assigned to the wrong category after the best matching of cluster labels to categories. To judge how far a method is from chance, we permute the true labels and recompute the matched error; across all conditions this random-label reference lies between 62.9% and 63.5%. All intervals are 95% intervals for a mean over 100 independent data sets. Method comparisons are paired within data sets and are not adjusted for multiple comparisons.

4 Results

4.1 Matched-total design at $r = 2$

Figure 3 summarizes the results at $r = 2$. When active users are focused (left column), concentrating activity helps Count and Jaccard but hurts Weighted and SVD. At $m = 12$, Count error falls from 35.5% under equal activity to 22.4% and then 13.0% as activity becomes more concentrated, while Weighted rises from 35.3% to 39.1% and 48.6%. Heavy buyers who stay within their category are the most informative users in the data, and the count projection lets them speak loudest.

At $m = 6$ the pattern is weaker. At high heterogeneity Count and Weighted coincide (41.3% and 41.2%): the 80% of users with one purchase contribute no pairs, so every contributing user buys 26 products and the two weightings differ by a constant.

When the active users are broad (right column), recovery is poor under medium and high

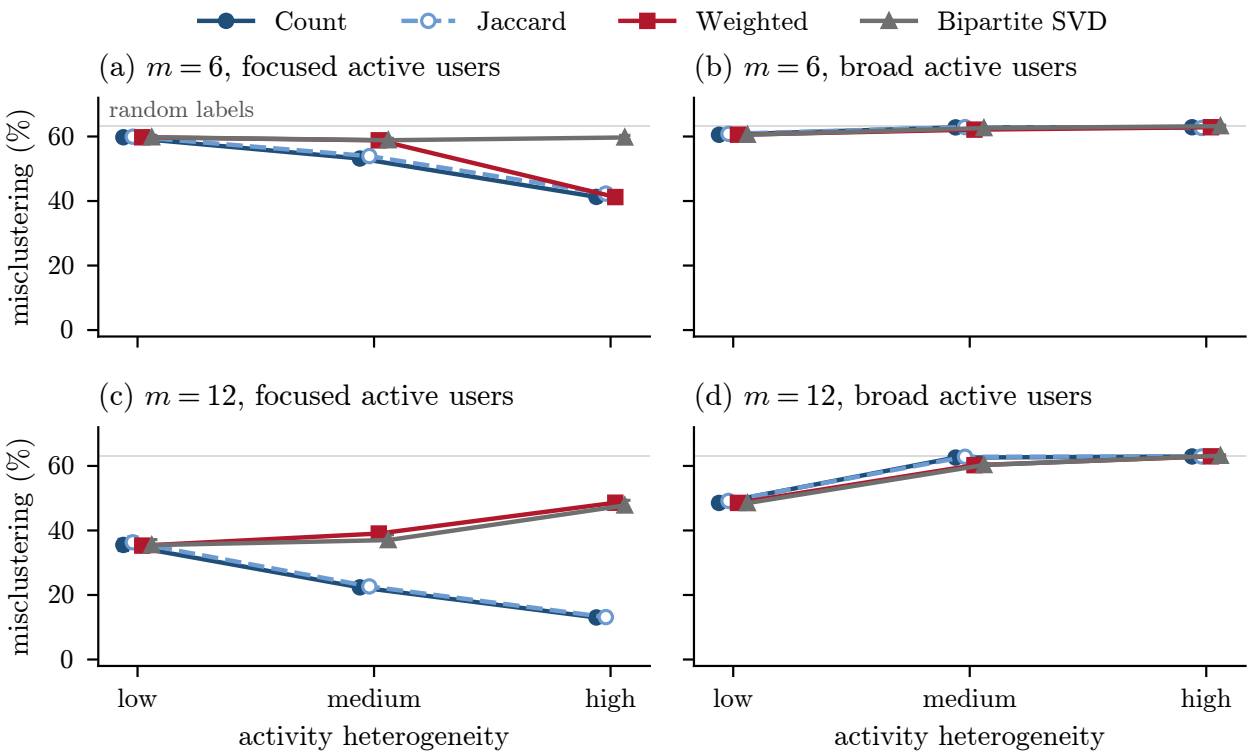


Figure 3: Mean misclustering rate at $r = 2$. Each point averages 100 data sets; error bars are 95% intervals and are often smaller than the markers. The grey band marks the random-label reference.

heterogeneity: every method is within about three points of the random-label reference. With equal activity it is better (48.5% at $m = 12$) because the broad users are no more active than anyone else.

4.2 The paired effect of weighting

Comparing Count and Weighted on the same data sets isolates the effect of weighting (all 12 paired differences at $r = 2$ are shown in Figure A2). With focused active users, weighting costs 16.7 percentage points at $m = 12$ and medium heterogeneity (95% interval 15.2 to 18.2) and 35.6 points at high heterogeneity (34.1 to 37.0). Its only clear gains occur with broad active users at medium heterogeneity, 2.4 points at $m = 12$ (1.8 to 3.0) and 0.7 points at $m = 6$, and both leave the error close to chance. Under equal activity the two networks are proportional, so their normalized matrices are identical and the small differences come from k-means initialization.

Why does weighting not rescue the broad-active conditions? At $m = 6$ and high heterogeneity, the 800 users with preferences buy one product each and create no pairs, while the 200 heavy buyers have no preference. The expected block contrast is exactly zero under every projection, and no reweighting can create signal that is not there. At $m = 12$, the focused users buy two products each and the broad users 52, so a broad user contributes 1,326 pairs under Count and 26 under Weighted, against one for a focused user. Weighting lowers the broad users' share of pair weight from 99.7% to 86.7%, but they still dominate. Because total

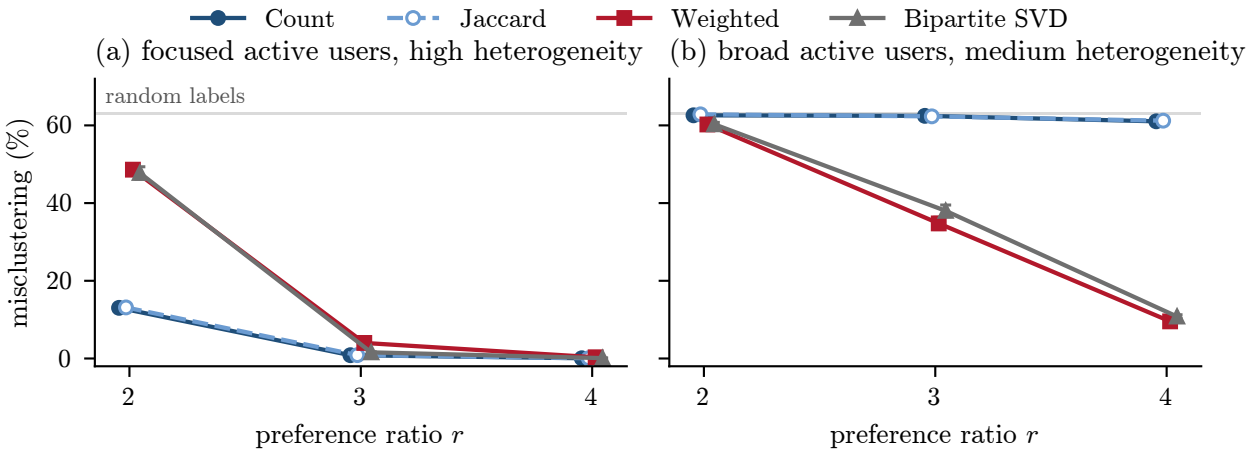


Figure 4: Misclustering rate as preference strength increases, at $m = 12$. Error bars are 95% intervals.

purchases are matched, making active users broad also takes purchases away from informative users, which is the confound the fixed-information design removes (Section 4.4).

4.3 Stronger preferences

Raising the preference strength changes the picture (Figure 4; complete results in Section D). With broad active users at $m = 12$ and medium heterogeneity, the methods now separate. COUNT and JACCARD stay at chance, while WEIGHTED falls to 34.8% at $r = 3$ and 9.6% at $r = 4$. The paired advantage of weighting is 27.7 points (26.1 to 29.3) and 51.5 points (50.8 to 52.2). In the focused conditions, stronger preferences make recovery easy for every method, so the cost of weighting shrinks; at $r = 4$ all three projections stay below 5.4% error. At high heterogeneity the broad-active conditions remain at chance for the reason given above.

4.4 Adding background to fixed informative purchases

Holding the informative purchases fixed separates the two effects (Figure 5). In the first setting, 50 background users buying $4m = 48$ products each push COUNT from 35.5% to 59.1%, whereas WEIGHTED stays at 36.0%; the paired difference is 23.0 points (21.4 to 24.7). In the second setting the difference at $D = 4m$ is 35.2 points (33.9 to 36.5). When $D = m$ the background users are no more active than anyone else, weighting cannot single them out, and the two methods give identical results. Weighting does not make recovery immune to background: with 200 background users, WEIGHTED also loses a few points.

4.5 Jaccard and the bipartite SVD

COUNT and JACCARD are nearly interchangeable here: their edge weights have a mean Spearman correlation of 0.97 and their mean errors never differ by more than 1.0 point, although their partitions still differ (mean adjusted Rand index 0.70). This is expected when

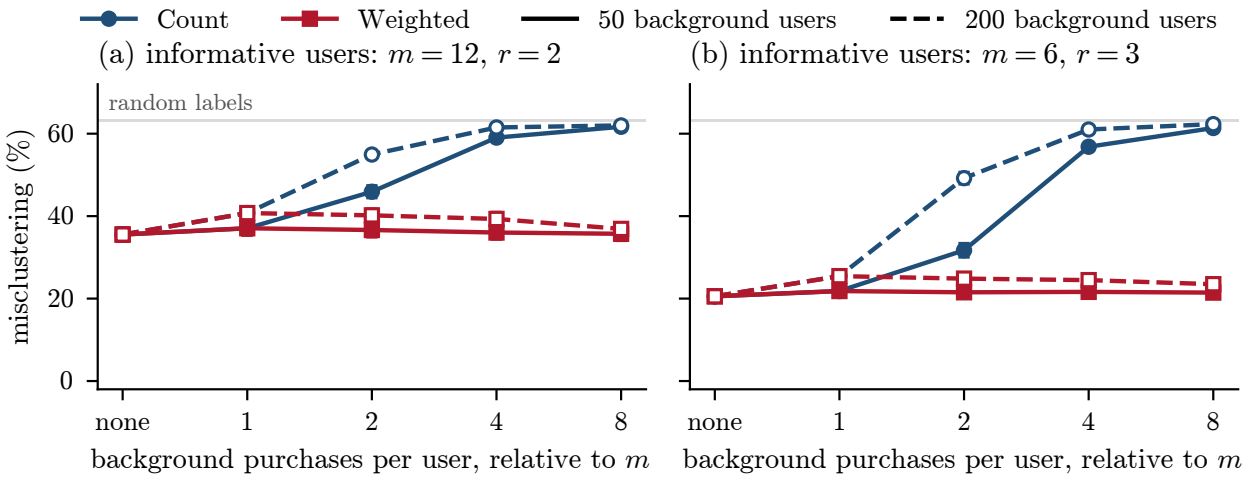


Figure 5: Adding preference-free users to 1,000 fixed informative users: solid lines add 50, dashed lines 200. JACCARD behaves like COUNT and SVD like WEIGHTED (Section D).

Table 2: SVD with and without single-purchase users (focused users, high heterogeneity). Mean error (%) over 100 recorded data sets; paired reduction with 95% interval.

Setting	COUNT	WEIGHTED	SVD	SVD, single-purchase users removed	Reduction (points)
$m = 6$, $r = 2$	41.3	41.2	59.7	41.3	18.4 (17.0, 19.9)
$m = 6$, $r = 3$	7.6	7.6	39.3	7.6	31.7 (29.4, 34.0)
$m = 12$, $r = 2$	13.0	48.6	47.8	47.8	0.0

all products in a category are equally popular.

The bipartite SVD tracks WEIGHTED, as (1) predicts, and offers no general advantage: with focused users at $m = 12$ and high heterogeneity its error exceeds that of COUNT by 34.7 points (33.2 to 36.2), and at $m = 6$ it trails both projections (59.7% against 41%).

Why does the SVD fail at $m = 6$? Users with one purchase contribute nothing to any projection but still enter the SVD through D_u, D_p, and the diagonal of $M^\top M$. In a follow-up check, rerunning the SVD on the recorded data sets without them (after checking that the original errors are reproduced exactly) brings it in line with COUNT at $m = 6$; at $m = 12$, where no user buys a single product, nothing changes (Table 2).

5 Discussion

Q1. Activity heterogeneity amplifies whatever active users do: concentrated focused activity strengthens the signal, while concentrated broad activity adds noise and removes informative purchases.

Q2. Inverse-activity weighting helps only when high activity marks uninformative users. With focused active users it is costly, by up to 36 points at $r = 2$. With broad active users it helps once the remaining users carry enough signal: by 28 and 52 points at $r = 3$ and $r = 4$,

and by up to 40 points when informative purchases are held fixed, while raw counts collapse to chance.

Q3. The normalized bipartite SVD behaves much like the weighted projection, as (1) predicts, offers no general advantage, and is hurt by single-purchase users.

Limitations. Our model is deliberately simple: balanced categories, equally popular products within a category, single-category preferences, and known K. Projected edges are dependent, so we do not use detection thresholds for independent-edge models. The fixed-information design was added after the matched-total results at $r = 2$ revealed the confound. We did not study $r < 2$ or real purchase data.

Conclusion. Activity is not a proxy for noise: downweighting heavy buyers helps when they buy at random and hurts when they are focused.

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A Exact category moments

Fix a user with d purchases and preference ratio r, and let $h = p - s$ be the number of non-preferred products. Let $P_t(c)$ be the probability that c of the first t purchases fall in the preferred category, with $P_0(0) = 1$. Given state c after t draws, the next product is preferred with probability

$$a_t(c) = \frac{r(s-c)}{r(s-c) + h - (t-c)},$$

so $P_{t+1}(c+1)$ += $P_t(c)\, a_t(c)$ and $P_{t+1}(c)$ += $P_t(c)\,[1 - a_t(c)]$. After d steps let C be the preferred count and $R = d - C$. Given C, the non-preferred purchases are a uniform R-subset of the h non-preferred products, so the expected number of ordered same-category pairs is

$$A = \mathbb{E}[C(C-1)] + (K-1)\,\mathbb{E}[R(R-1)]\,\frac{s(s-1)}{h(h-1)}.$$

Averaging over a uniformly drawn preferred category gives $q^{\text{same}} = A/[Ks(s-1)]$ and $q^{\text{diff}} = [d(d-1) - A]/[K(K-1)s^2]$.

B Agreement between theory and simulation

Figure A1 compares the expected within-minus-between contrast with its simulated mean in each of the 12 main conditions. For example, at $m = 12$ with focused users and high heterogeneity the expected Count contrast is 0.902 and the simulated mean is 0.908; for Weighted the values are 0.0210 and 0.0212.

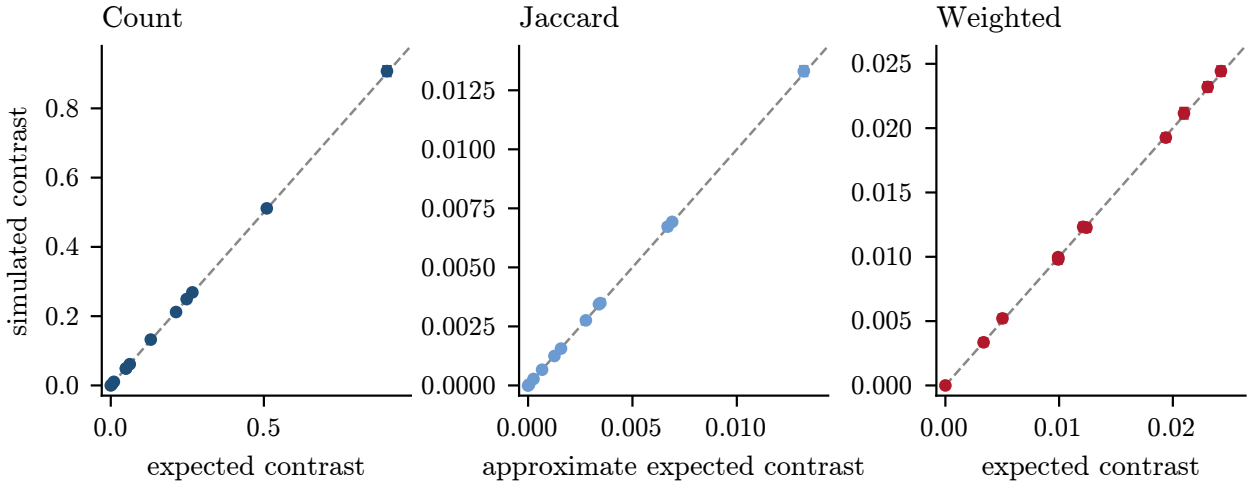


Figure A1: Expected versus simulated within-minus-between contrast, one point per condition, with 95% intervals. The dashed line marks equality. Each panel uses its own units; Jaccard uses the ratio-of-expectations approximation.

C Reproducibility

All simulations use Python 3.11 with the package versions recorded in `requirements-lock.txt`. Root seeds are 23601 (pilot), 23603 ($r = 2$), 23604 ($r = 3$), 23605 (fixed-information design), 23607 ($r = 4$), and 23606 (label permutations). Every data set's seed and k-means seed is stored, so any purchase matrix can be regenerated without storing it. In the $r = 3$, $r = 4$, and fixed-information data sets all methods share one k-means seed, which makes the exact equivalence of Count and Weighted under equal activity visible in the results. Unit tests check the design identities, label matching, recovery of a known partition, and the theoretical moments against simulation; validation scripts check every saved summary and paired interval. Mean runtimes per data set were 12.5 ms for Count, 12.8 ms for Jaccard, 14.8 ms for Weighted, and 48.2 ms for SVD, excluding data generation. The repository, `https://github.com/diaomj668/236project`, lists the commands that reproduce every result and figure in this report.

D Complete results

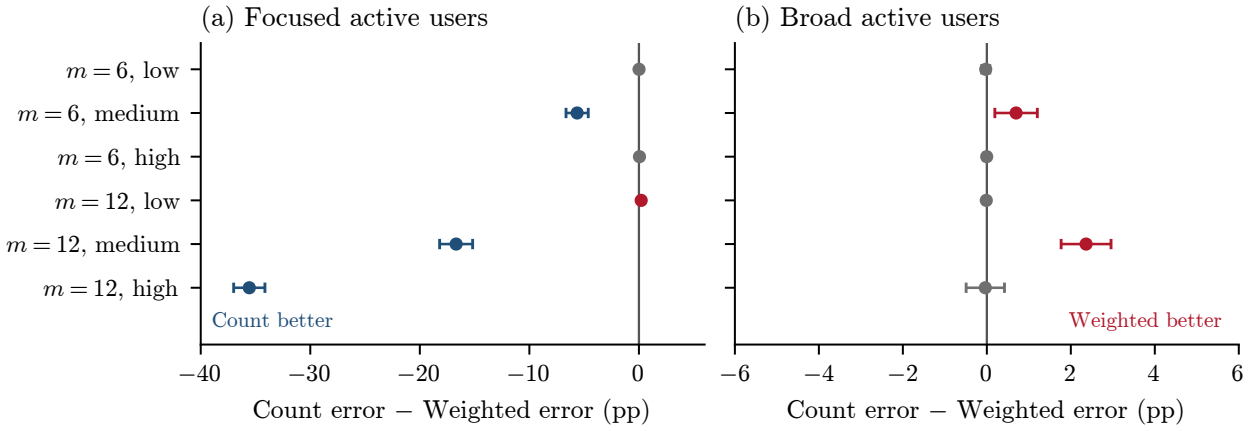


Figure A2: Paired difference in misclustering rate, Count minus Weighted, with 95% intervals over 100 data sets. Negative values favour Count. Note the different horizontal scales.

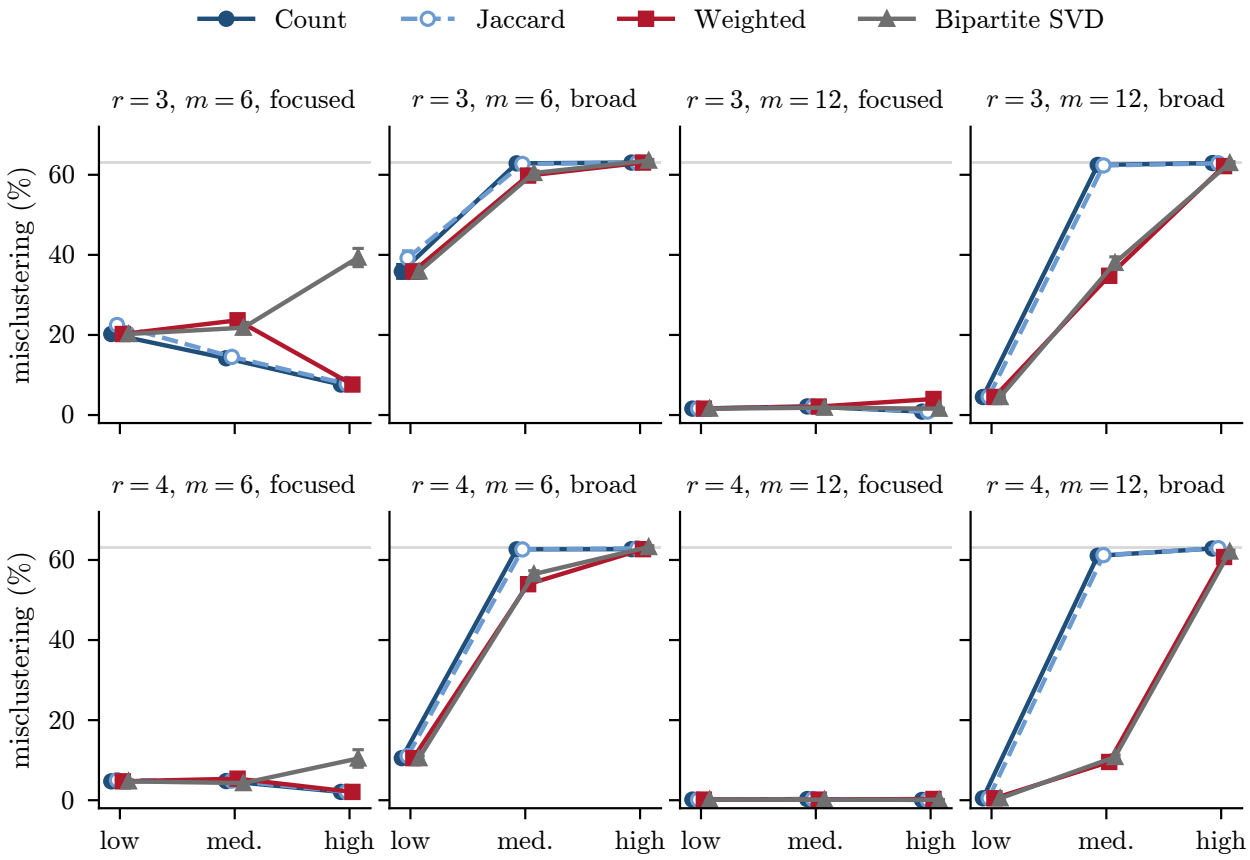


Figure A3: Results at $r = 3$ (top) and $r = 4$ (bottom), with the same layout as Figure 3. The grey band marks the random-label reference.

Table A1: Results at $r = 2$: mean misclustering rate (%) over 100 data sets per condition.

m	Heterogeneity	Active users	Count	Jaccard	Weighted	SVD
6	low	focused	59.8	59.9	59.8	59.8
6	low	broad	60.5	60.8	60.6	60.5
6	medium	focused	53.2	53.9	58.8	58.8
6	medium	broad	62.8	62.9	62.1	62.6
6	high	focused	41.3	42.3	41.2	59.7
6	high	broad	62.9	62.6	62.9	63.2
12	low	focused	35.5	36.3	35.3	35.5
12	low	broad	48.5	49.2	48.5	48.5
12	medium	focused	22.4	22.7	39.0	37.0
12	medium	broad	62.6	62.8	60.2	60.3
12	high	focused	13.0	13.1	48.6	47.8
12	high	broad	62.9	62.9	63.0	63.2

Table A2: Results at $r = 3$: mean misclustering rate (%).

m	Heterogeneity	Active users	COUNT	JACCARD	WEIGHTED	SVD
6	low	focused	20.2	22.5	20.2	20.2
6	low	broad	35.9	39.2	35.9	35.9
6	medium	focused	14.2	14.5	23.7	21.8
6	medium	broad	62.9	62.7	59.8	60.5
6	high	focused	7.6	7.8	7.6	39.3
6	high	broad	63.1	63.0	63.1	63.5
12	low	focused	1.6	1.6	1.6	1.6
12	low	broad	4.5	4.6	4.5	4.5
12	medium	focused	2.1	2.2	2.1	1.8
12	medium	broad	62.5	62.3	34.8	38.0
12	high	focused	0.9	0.9	4.0	1.6
12	high	broad	62.9	62.9	62.2	63.0

Table A3: Results at $r = 4$: mean misclustering rate (%).

m	Heterogeneity	Active users	COUNT	JACCARD	WEIGHTED	SVD
6	low	focused	4.7	5.0	4.7	4.7
6	low	broad	10.5	11.0	10.5	10.5
6	medium	focused	4.8	4.9	5.4	4.3
6	medium	broad	62.7	62.6	53.9	56.5
6	high	focused	2.1	2.1	2.1	10.4
6	high	broad	62.7	62.9	62.7	63.2
12	low	focused	0.1	0.1	0.1	0.1
12	low	broad	0.5	0.5	0.5	0.5
12	medium	focused	0.3	0.3	0.1	0.1
12	medium	broad	61.0	61.2	9.6	10.8
12	high	focused	0.1	0.1	0.3	0.1
12	high	broad	62.9	62.9	60.7	62.1

Table A4: Fixed-information design: mean misclustering rate (%) over 100 data sets. A is the number of added preference-free users and D/m their purchases relative to the informative users.

Informative users	A	D/m	Count	Jaccard	Weighted	SVD
$m = 12$, $r = 2$	0	–	35.5	36.4	35.5	35.5
	50	1	37.1	37.4	37.1	37.1
	50	2	45.9	46.3	36.6	36.8
	50	4	59.0	59.2	36.0	36.3
	50	8	61.7	61.8	35.7	36.1
	200	1	40.7	41.5	40.7	40.7
	200	2	54.9	55.2	40.2	40.3
	200	4	61.5	61.4	39.3	39.8
	200	8	62.0	62.0	36.9	38.3
$m = 6$, $r = 3$	0	–	20.5	22.3	20.5	20.5
	50	1	21.8	23.5	21.8	21.8
	50	2	31.7	34.4	21.5	21.8
	50	4	56.8	57.3	21.6	22.3
	50	8	61.4	61.6	21.4	22.7
	200	1	25.4	27.3	25.4	25.4
	200	2	49.2	50.2	24.8	26.2
	200	4	61.0	60.9	24.5	27.0
	200	8	62.3	62.4	23.5	26.4